Linear equations in one variable — Part 2

Algebra · Math · 56 questions

2.1 7084509a-2eca-4996-a36e-1e7ed19fdd12 · Easy

$$4x + 6 = 18$$

Which equation has the same solution as the given equation?

A. $4x = 108$

B. $4x = 24$

C. $4x = 12$

D. $4x = 3$

2.2 027085-DC · Easy

$6x + k = 6x + 5$

In the given equation, k is a constant. If the equation has infinitely many solutions, what is the value of k ?

Student-produced response: _____

2.3 f3aafaa2-b75f-4b93-a443-b88b79f2a8b1 · Easy

If $4x + 2 = 12$, what is the value of $16x + 8$?

A. 40

B. 48

C. 56

D. 60

2.4 06911-DC · Easy

Cathy has n CDs. Gerry has 3 more than twice the number of CDs that Cathy has. In terms of n, how many CDs does Gerry have?

A. $3n-2$

B. $3n+2$

C. $2n-3$

D. $2n+3$

2.5 55b379da-d6e2-4c9e-857d-8da822c1d1fa · Easy

$$8x - 7x + 130 = 260$$

What value of x is the solution to the given equation?

Student-produced response: _____

2.6 4d9de101-04d3-4a9b-96ae-eef85f72bd23 · Easy

Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid $\$ 2$ for the box of cereal and $\$ 1.90$ per pound for the strawberries. If Lorenzo paid a total of $\$ 9.60$ for the box of cereal and the strawberries, which of the following equations can be used to find p, the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

A. $1.90p + 2 = 9.60$

B. $1.90p - 2 = 9.60$

C. $1.90 + 2p = 9.60$

D. $1.90 - 2p = 9.60$

2.7 ad73f33e-e2f7-442f-9463-281f925a0111 · Easy

If $2x + 3 = 9$, what is the value of $6x - 1$?

Student-produced response: _____

2.8 79f6861f-e1b2-4b3a-a56b-065a572689b2 · Easy

A rocket contained $467{,}000$ kilograms (kg) of propellant before launch. Exactly 21 seconds after launch, $362{,}105$ kg of this propellant remained. On average, approximately how much propellant, in kg, did the rocket burn each second after launch?

A. $4{,}995$

B. $17{,}243$

C. $39{,}481$

D. $104{,}895$

2.9 85822077-c33d-45ef-bd16-1f8decf34dde · Medium

A bowl contains 20 ounces of water. When the bowl is uncovered, the amount of water in the bowl decreases by 1 ounce every 4 days. If 9 ounces of water remain in this bowl, for how many days has it been uncovered?

A. 3

B. 7

C. 36

D. 44

2.10 070610-DC · Medium

$$2(p+1)+8(p-1)=5p$$

What value of p is the solution of the equation above?

Student-produced response: _____

2.11 565e2580-0194-4be0-b5a7-5e2273ce4a14 · Medium

$$66x = 66x$$

How many solutions does the given equation have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

2.12 657150dc-4949-4ed3-afe8-a00d56bf8642 · Medium

The cost to rent a bus from Company X is $\$ 950$ for the first 3 hours and an additional $\$ 50$ per hour for each hour after the first 3 hours. If the total cost to rent the bus from Company X for t hours, where $t > 3$, is $\$ 1{,}150$, which equation represents this situation?

A. $950(t - 3) + 50t = 1{,}150$

B. $950(3t) + 50t = 1{,}150$

C. $950 + 50(t - 3) = 1{,}150$

D. $950 + 50(3t) = 1{,}150$

2.13 11138-DC · Medium

The width of a rectangular dance floor is w feet. The length of the floor is 6 feet longer than its width. Which of the following expresses the perimeter, in feet, of the dance floor in terms of w ?

A. $2w + 6$

B. $4w + 12$

C. $w^2 + 6$

D. $w^2 + 6w$

2.14 028955-DC · Medium

$3(2x-6)-11=4(x-3)+6$

If x is the solution to the equation above, what is the value of $x-3$?

A. $\frac{23}{2}$

B. $\frac{17}{2}$

C. $\frac{15}{2}$

D. $-\frac{15}{2}$

2.15 ed19c902-acac-433c-b8f5-c8cd07c4e16d · Medium

A candle is made of 17 ounces of wax. When the candle is burning, the amount of wax in the candle decreases by 1 ounce every 4 hours. If 6 ounces of wax remain in this candle, for how many hours has it been burning?

A. 3

B. 6

C. 24

D. 44

2.16 8e8223b7-2824-48a9-a122-1abf78976190 · Medium

A company that creates and sells tape dispensers calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. The equation $15{,}000 = 2.00x - 4{,}500$ represents this situation for a month where x tape dispensers are created and sold. Which statement is the best interpretation of $2.00x$ in this context?

A. The monthly sales revenue, in dollars, from selling x tape dispensers

B. The monthly sales revenue, in dollars, from each tape dispenser sold

C. The monthly cost, in dollars, of creating each tape dispenser

D. The monthly cost, in dollars, of creating x tape dispensers

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2.17 027743-DC · Medium

$(b-2)x=8$

In the given equation, b is a constant. If the equation has no solution, what is the value of b ?

A. 2

B. 4

C. 6

D. 10

2.18 a24605a1-dcc2-4d33-8f1f-a5e3f3904603 · Medium

$$2x + 16 = ax + 8$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

Student-produced response: _____

2.19 0964937e-bca9-49a1-ae3a-255505993ab8 · Medium

If $46 = 16 + 2x - 8$, what is the value of $2x - 8$?

A. 16

B. 23

C. 30

D. 38

2.20 05043-DC · Medium

Megan's regular wage at her job is p dollars per hour for the first 8 hours of work in a day plus 1.5 times her regular hourly wage for work in excess of 8 hours that day. On a given day, Megan worked for 10 hours, and her total earnings for that day were $137.50. What is Megan's regular hourly wage?

A. $11.75

B. $12.50

C. $13.25

D. $13.75

2.21 08357-DC · Medium

If $2(x-5)+3(x-5)=10$, what is the value of $x-5$?

A. 2

B. 5

C. 7

D. 12

2.22 08050-DC · Medium

$a(3-x)-b=-1-2x$

In the equation above, a and b are constants. If the equation has infinitely many solutions, what are the values of a and b ?

A. $a=2$ and $b=1$

B. $a=2$ and $b=7$

C. $a=-2$ and $b=5$

D. $a=-2$ and $b=-5$

2.23 81094fb4-28a4-4d27-b028-07b229772468 · Medium

The cost to rent a commercial fishing boat from a certain company is $\$ 950$ for the first 2 hours and an additional $\$ 50$ per hour for each hour after the first 2 hours. If the total cost to rent the commercial fishing boat from the company for t hours, where $t > 2$, is $\$ 1{,}100$, which equation represents this situation?

A. $950(t - 2) + 50t = 1{,}100$

B. $950(2t) + 50t = 1{,}100$

C. $950 + 50(t - 2) = 1{,}100$

D. $950 + 50(2t) = 1{,}100$

2.24 f0727bf9-4765-47a5-ad90-ee9cc0ed3794 · Medium

If $25x = 6$, what is the value of $5x$?

A. $\frac{4}{5}$

B. $\frac{5}{3}$

C. 3

D. 4

2.25 ff27b3f5-a00f-4258-b4c6-8f41c7c06bf0 · Medium

If $5x + 4 = 4x + 4 + 29$, what is the value of $x + 4$?

A. -4

B. 25

C. 29

D. 33

2.26 162bcb97-414c-46ed-a490-4d88402f07fa · Medium

$$4x + 12 = \frac{ax + 3}{2}$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

A. 0

B. 3

C. 8

D. 12

2.27 028876-DC · Medium

$2n + 6 = 14$

A tree had a height of 6 feet when it was planted. The equation above can be used to find how many years n it took the tree to reach a height of 14 feet. Which of the following is the best interpretation of the number 2 in this context?

A. The number of years it took the tree to double its height

B. The average number of feet that the tree grew per year

C. The height, in feet, of the tree when the tree was 1 year old

D. The average number of years it takes similar trees to grow 14 feet

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2.28 4f88a160-8333-4005-8a92-1ba7b2c74b99 · Medium

$$\frac{1}{3}x + 6 - \frac{1}{2}x + 6 = -8$$

What value of x is the solution to the given equation?

Student-produced response: _____

2.29 ba23c485-785e-419e-b55c-cb7482fba61e · Medium

If $2(3t - 10) + t = 40 + 4t$, what is the value of $3t$?

Student-produced response: _____

2.30 022123-DC · Medium

If $4x - \frac{1}{2} = -5$, what is the value of $8x - 1$?

A. 2

B. $-\frac{9}{8}$

C. $-\frac{5}{2}$

D. -10

2.31 568627eb-0784-4a63-a56a-ddffad314bb8 · Medium

If $\frac{6}{7}p + 18 = 54$, what is the value of $7p$?

Student-produced response: _____

2.32 63a0e21a-3df7-463a-b0b7-6bf2eeefbcb4 · Medium

$$\frac{1}{4}x + 5 - \frac{1}{3}x + 5 = -7$$

What value of x is the solution to the given equation?

A. -12

B. -5

C. 79

D. 204

2.33 6da250b9-a95c-48c8-b785-d54366f667e3 · Medium

$$9r = 7(r + 7)$$

What value of r is the solution to the given equation?

Student-produced response: _____

2.34 d92e58bf-e6da-484f-8d84-5452a09619b1 · Medium

If $9(4 - 3x) + 2 = 8(4 - 3x) + 18$, what is the value of $4 - 3x$?

A. -16

B. -4

C. 4

D. 16

2.35 070609-DC · Medium

An agricultural scientist studying the growth of corn plants recorded the height of a corn plant at the beginning of a study and the height of the plant each day for the next 12 days. The scientist found that the height of the plant increased by an average of 1.20 centimeters per day for the 12 days. If the height of the plant on the last day of the study was 36.8 centimeters, what was the height, in centimeters, of the corn plant at the beginning of the study?

Student-produced response: _____

2.36 4b647461-1e29-465f-a480-e7b4dcecc200 · Medium

A museum rents tablets to visitors. The museum earns revenue of $\$ 14$ for each tablet rented for the day. On Wednesday, the museum earned $\$ 406$ in profit from renting tablets after paying daily expenses of $\$ 112$. How many tablets did the museum rent on Wednesday? $\text{profit} = \text{total revenue} - \text{total expenses}$

Student-produced response: _____

2.37 e2acc363-d0b7-4b19-ac43-34982f030a3d · Hard

Each side of a 30-sided polygon has one of three lengths. The number of sides with length $8 \text{ centimeters } \text{cm}$ is 5 times the number of sides n with length 3 cm. There are 6 sides with length 4 cm. Which equation must be true for the value of n?

A. $5n + 6 = 30$

B. $6n + 6 = 30$

C. $8n + 3n + 4n = 30$

D. $85n + 3n + 46 = 30$

2.38 10864-DC · Hard

The equation $9x+5=a(x+b)$, where a and b are constants, has no solutions. Which of the following must be true?

I. $a=9$

II. $b=5$

III. $b \neq \frac{5}{9}$

A. None

B. I only

C. I and II only

D. I and III only

2.39 c17521a8-2834-4b36-9f1a-dca9bf6e906d · Hard

$$2kx - n = -\frac{28}{15}x - \frac{36}{19}$$

In the given equation, k and n are constants and $n > 1$. The equation has no solution. What is the value of k?

Student-produced response: _____

2.40 98dc96db-8257-4f49-95af-515712d5e4e3 · Hard

How many solutions does the equation $1015x - 9 = -156 - 10x$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

2.41 033d8e6a-f7e5-456a-b3ab-38b7ad87a27d · Hard

A factory makes 9-inch, 7-inch, and 4-inch concrete screws. During a certain day, the number of 9-inch concrete screws that the factory makes is 5 times the number n of 7-inch concrete screws, and the number of 4-inch concrete screws is 22. During this day, the factory makes 100 concrete screws total. Which equation represents this situation?

A. $95n + 7n + 422 = 100$

B. $9n + 7n + 4n = 100$

C. $5n + 22 = 100$

D. $6n + 22 = 100$

2.42 2a3b6ab6-2f25-4e2a-817b-9112e317c958 · Hard

$$x(r - 9) + 4 = 19x + 27$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r?

A. 4

B. 9

C. 23

D. 28

2.43 6d24c78b-537d-4e65-8a41-211301b6c972 · Hard

How many solutions does the equation $12x - 3 = -3x + 12$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

2.44 9d5bc04d-81bd-4fda-8c78-8c26fa0de2f0 · Hard

$$-3x + 21px = 84$$

In the given equation, p is a constant. The equation has no solution. What is the value of p?

A. 0

B. $\frac{1}{7}$

C. $\frac{4}{3}$

D. 4

2.45 c5442207-cd62-4203-8c5a-95d42e060e41 · Hard

$$-49x = -98x$$

How many solutions does the given equation have?

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

2.46 7c776d67-dc96-4c86-97df-52e3204cdfe9 · Hard

What value of t is the solution to the equation $0.8t - 0.46 = 8t - 0.001 + 1.9$?

Student-produced response: _____

2.47 d406296c-cba6-440d-94d6-5bf2accfba2a · Hard

$$5(t + 3) - 7(t + 3) = 38$$

What value of t is the solution to the given equation?

Student-produced response: _____

2.48 05397-DC · Hard

Hector used a tool called an auger to remove corn from a storage bin at a constant rate. The bin contained 24,000 bushels of corn when Hector began to use the auger. After 5 hours of using the auger, 19,350 bushels of corn remained in the bin. If the auger continues to remove corn at this rate, what is the total number of hours Hector will have been using the auger when 12,840 bushels of corn remain in the bin?

A. 3

B. 7

C. 8

D. 12

2.49 0797e1a5-ab62-4b2e-b8db-ec842244529a · Hard

If $5 - 72 - 4x = 16 - 82 - 4x$, what is the value of $2 - 4x$?

Student-produced response: _____

2.50 08060-DC · Hard

A science teacher is preparing the 5 stations of a science laboratory. Each station will have either Experiment A materials or Experiment B materials, but not both. Experiment A requires 6 teaspoons of salt, and Experiment B requires 4 teaspoons of salt. If x is the number of stations that will be set up for Experiment A and the remaining stations will be set up for Experiment B, which of the following expressions represents the total number of teaspoons of salt required?

A. $5x$

B. $10x$

C. $2x + 20$

D. $10x + 20$

2.51 ad6cdfd4-4643-45b2-b72d-0602fb30ce1c · Hard

$$\frac{12x+28}{4} - \frac{s}{13} = rx - 8$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s?

Student-produced response: _____

2.52 830cf5a2-7b06-43e1-bf6c-fb7814a69db0 · Hard

If $\frac{x+6}{3} = \frac{x+6}{13}$, the value of $x + 6$ is between which of the following pairs of values?

A. -7 and -3

B. -2 and 2

C. 2 and 7

D. 8 and 13

2.53 05364-DC · Hard

Alan drives an average of 100 miles each week. His car can travel an average of 25 miles per gallon of gasoline. Alan would like to reduce his weekly expenditure on gasoline by $5. Assuming gasoline costs $4 per gallon, which equation can Alan use to determine how many fewer average miles, m, he should drive each week?

A. $\frac{25}{4}m = 95$

B. $\frac{25}{4}m = 5$

C. $\frac{4}{25}m = 95$

D. $\frac{4}{25}m = 5$

2.54 e9630635-49ff-4317-9dc7-81fb2b4d2e19 · Hard

$$3(kx + 13) = \frac{48}{17}x + 36$$

In the given equation, k is a constant. The equation has no solution. What is the value of k?

Student-produced response: _____

2.55 eec9f1c3-9c60-420a-98bf-68fc90a14f95 · Hard

If $\frac{x - 5}{7} = \frac{x - 5}{9}$, the value of $x - 5$ is between which of the following pairs of values?

A. -9 and -7

B. -3 and 3

C. 4.5 and 5.5

D. 6.75 and 9.25

2.56 0d482312-5261-43da-8a42-d4ab593d0949 · Hard

A manufacturing plant makes 10-inch, 9-inch, and 7-inch frying pans. During a certain day, the number of 10-inch frying pans that the manufacturing plant makes is 4 times the number n of 9-inch frying pans it makes, and the number of 7-inch frying pans it makes is 10. During this day, the manufacturing plant makes 100 frying pans total. Which equation represents this situation?

A. $104n + 9n + 710 = 100$

B. $10n + 9n + 7n = 100$

C. $4n + 10 = 100$

D. $5n + 10 = 100$
